Analysis on the Application of Machine Learning Models and Their Interpretability in Credit Default Risk Prediction

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***Abstract*—This project examines the use of machine learning models to predict customer defaults and compares the performance of baseline model with advanced models. While advanced models showed slight improvements in Recall and AUC, the differences were minimal, likely due to a limited feature set and the need for further hyper-parameter tuning. Using SHAP values, key predictors such as age, employment stability, loan amount, and spending behavior were identified, providing valuable insights into default risk. The findings highlight the importance of explainable AI tools and data preprocessing in building reliable credit risk prediction models.**

***Index Terms*—credit risk, machine learning, deep learning, model interpretability**

I. Introduction

A. Background and Motivation

With the rapid growth of the credit market, it has become increasingly vital for banks and lending institutions to identify potential default risks that could result in significant financial losses. Given the vast amounts of customer data collected, machine learning models have emerged as essential tools for predicting defaults, providing valuable insights, and formulating risk management strategies.

B. Project Goal

This project aims to evaluate the performance of various machine learning models in detecting customer defaults, enabling better risk management and decision-making.

C. Literature Review

Statistical models have long been foundational in credit risk management, with logistic regression standing out for its simplicity and interpretability. Seminal works by Ohlson [1] and Altman [2] emphasized the use of logistic regression and discriminant analysis in credit scoring. However, these linear models often struggled to capture complex, non-linear relationships in data, prompting the adoption of machine learning (ML) methods. Breiman's [3] Random Forest improved predictive accuracy through bagging, while Chen and Guestrin [4] highlighted the robustness of XGBoost, a gradient boosting method, in handling diverse datasets, including credit risk.

In recent years, deep learning (DL) models have gained prominence, driven by the rise of big data and advances in computational power. Studies like Das and Huang [5] have explored recurrent neural networks (RNNs) for dynamic credit scoring, leveraging time-series features to enhance predictions. Despite their potential, DL models face significant challenges in practical applications, particularly their lack of interpretability, which remains critical for regulatory compliance in the financial industry.

II. Methods

A. Problem Formulation

In this project, two questions will be researched. First, do complex models, including ML and DL methods, significantly outperform logistic regression in credit default prediction? Second, what are the most influential features for predicting defaults, and how can these be effectively identified across both models?

The dataset used in this project was sourced from Hugging Face and originally provided by the Home Credit Group, an international credit company. It consists of 307,511 rows and 122 columns, containing both categorical and numerical data. Each row represents a customer's credit history record, including details such as customer ID, gender, occupation, annual income, family size, and more.

The target variable, TARGET, is binary and indicates whether a customer defaulted or not on their credit in the current month (1 for default, 0 for not). This makes it a binary classification problem. The class imbalance is notable, with a ratio of approximately 1:12 (Default: Not default), which suggests the need for resampling methods to balance the dataset.

B. Feature Selection

The original dataset includes 122 features, and lots of them are sparse. Therefore, it is necessary to eliminate those with minimal predictive power. In the credit risk management industry, Weight of Evidence (WOE) and Information Value (IV) are widely used to understand the predictive power of an independent variable. Let $P(\text{Non-event})$ and $P(\text{Event})$ represent the proportions of non-events (non-defaults) and events (defaults), respectively. The formula of WOE is:

$$\text{WOE} = \ln \frac{P(\text{Non-event})}{P(\text{Event})} \tag{1}$$

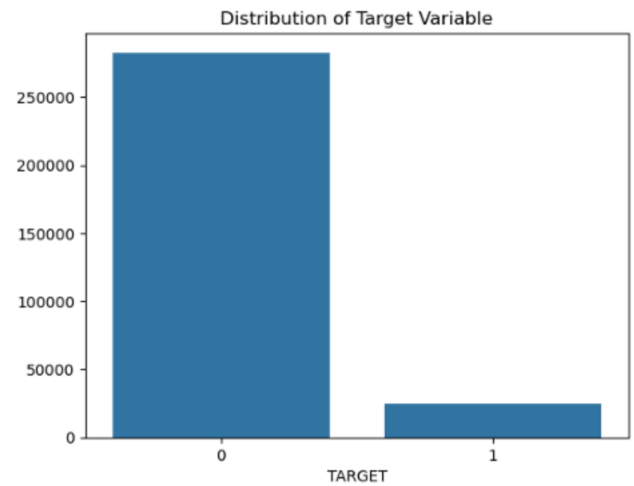

Fig. 1. Distribution of Target Variable Before Resampling.

and IV is calculated by:

$$\text{IV} = \sum \text{WOE} \cdot (P(\text{Non-event}) - P(\text{Event})) \tag{2}$$

Typically, predictors with an IV value below 0.02 are considered weak, while those with higher values are retained for further analysis. In this project, I set a stricter threshold of 0.03 to identify features with stronger predictive power. Using the `scorecardpy` package, I calculated the IV values for all features and selected the 32 features with relatively higher IV values for subsequent modeling.

C. Models

To evaluate the performance of various models, Logistic Regression (baseline), XGBoost, and Multilayer Perceptron (MLP) were selected. Logistic Regression and XGBoost have been widely used in credit scoring for years due to their effectiveness and interpretability. Neural network models, such as MLP, have seen limited adoption in this domain, primarily because of their lower interpretability. We will first use different metrics to evaluate the models' ability to distinguish between default and non-default cases. Additionally, we will analyze the features to identify those with the most significant influence in predicting defaults.

D. Metrics

In this project, Area Under the ROC Curve (AUC) and Recall are prioritized over Accuracy because they provide a more nuanced evaluation of model performance in imbalanced datasets. Accuracy can be misleading in scenarios where non-default cases significantly outnumber default cases, as a model could achieve high accuracy simply by predicting all cases as non-default. However, such a model would fail to identify actual default cases, which are critical in the credit risk management context.

AUC measures the model's ability to differentiate between classes across all thresholds. It is mathematically represented as:

$$\text{AUC} = \int_0^1 TPR(FPR)\, d(FPR) \tag{3}$$

where TPR is the true positive rate and FPR is the false positive rate.

Recall, on the other hand, evaluates the model's effectiveness in identifying default cases and is calculated as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \tag{4}$$

Higher Recall ensures that most default cases are correctly identified, reducing the risk of misclassification.

III. Results

A. Data Pipeline

The preprocessing steps include removal of extreme values, imputation of missing values using mode and median, one-hot encoding of categorical features, standardization, etc. After preprocessing and feature selection (stated before), the dataset contains 252,133 rows and 33 columns. The dataset was then randomly split into a training set (80%) and a test set (20%). To address data imbalance, the training set was resampled to ensure balanced representation, which is crucial in this context. After that, Logistic Regression, XGBoost Classifier, and Multilayer Perceptron models were developed and evaluated using AUC and recall metrics. Finally, feature importance was analyzed using SHAP plots and other feature importance tools to identify the key predictors.

B. Evaluation

TABLE I summarizes the performance of three models before resampling. It is evident that all models have low Recall scores (less than 0.5), highlighting their poor ability to identify default cases.

TABLE I
Model Performance Comparison before Resampling

Model	AUC	Recall
Logistic Regression	0.66490	0.34532
XGBoost Classifier	0.64574	0.32541
Multilayer Perceptron	0.65381	0.37144

TABLE II summarizes the performance of three models after resampling (class ratio 1:1). Each model demonstrates notable improvements in both Recall and AUC scores after applying a resampling method to the training set and refitting the models,indicating enhanced performance in detecting default cases. Take MLP model as an example, which has an AUC of 0.73825 and Recall of 0.67144. These scores indicate that the model has a 73% chance of correctly distinguishing between default and non-default cases, and it identifies 67% of actual default cases (true positives), which is acceptable in the context of credit risk prediction.

TABLE II
MODEL PERFORMANCE COMPARISON AFTER RESAMPLING

Model	AUC	Recall
Logistic Regression	0.73716	0.66437
XGBoost Classifier	0.70106	0.64131
Multilayer Perceptron	0.73825	0.67144

Although complex boosting and neural network models are often assumed to outperform linear models, I did not observe any significant improvement in prediction performance between the baseline and advanced models in this project. As shown in TABLE II, Logistic Regression and MLP have very similar AUC scores, with MLP only slightly outperforming Logistic Regression in terms of Recall. This outcome could be attributed to the limited number of features available after filtering out weak predictors or the need for further hyperparameter tuning in the neural network training process. Attempts to create a more complex MLP model with different hidden layer sizes resulted in overfitting, highlighting the challenges of balancing model complexity and generalization.

C. Feature Interpretation

In this project, Shapley values (SHAP) were used to understand feature importance. Figure 2 illustrates the SHAP values of each feature variable for all samples in the dataset, with each dot representing one sample from the dataset. The density of the dots corresponds to the distribution of SHAP values for that feature.

The X-axis represents the SHAP values, indicating whether the feature has a positive or negative impact on the model's final prediction. The Y-axis represents the distribution of feature values, where each cluster of dots for a feature variable reflects the distribution of that feature across all samples in the dataset. The color of the dots indicates the actual value of the feature, with the X-axis showing how each specific value affects the model's prediction direction and magnitude. Positive SHAP values push the model's prediction toward the positive class (e.g., higher risk of default) while negative SHAP values push the prediction toward the negative class (e.g., lower risk of default).

From Figure 2, we can see that "AMT GOODS PRICE", "AMT CREDIT", and "CODE GENDER M" are among the most influential predictors. High values of goods purchased and credit amount (red dots) are associated with increased risk of default, while older age ("DAYS BIRTH") and longer employment duration ("DAYS EMPLOYED") are associated with a reduced risk of default. In addition, "CODE GENDER M" indicates that being male (red dots) has a higher likelihood of influencing the model toward default compared to females. Findings from SHAP values actually match our expectation. Customers who are mature in age and hold stable jobs are generally more reliable for lending institutions, as they are typically better positioned to repay their loans on time. However, behaviors such as spending large amounts of money on shopping or borrowing significant sums can increase the risk of default, as these actions may reflect financial strain or poor money management, reducing their ability to meet repayment obligations. These insights underscore the importance of considering both demographic stability and financial behavior in credit risk assessments.

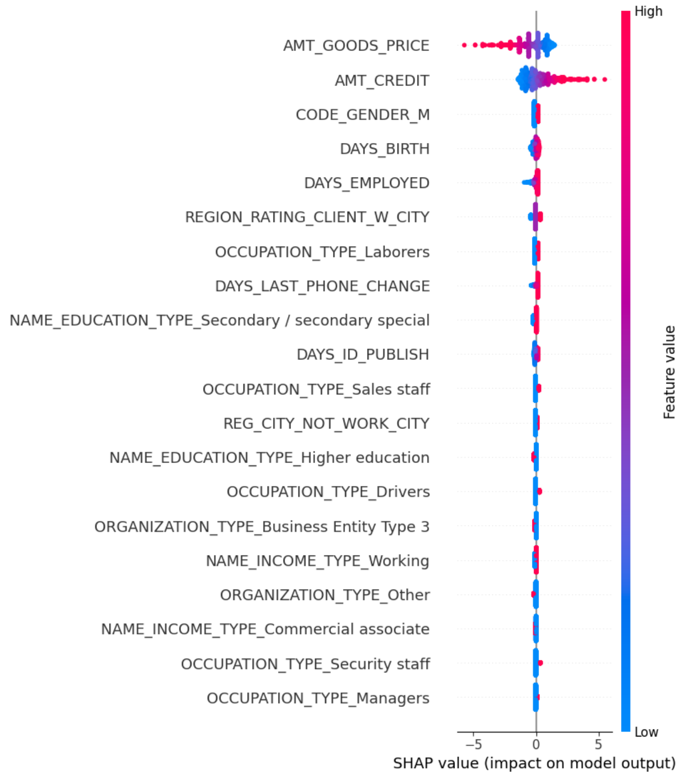

Fig. 2. SHAP Plot

CONCLUSIONS AND FUTURE WORK

In this project, I explored the application of machine learning models in predicting customer defaults, comparing the performance of baseline model Logistic Regression with more advanced models such as MLPClassifier and XGBoost. The results demonstrated that while complex models slightly outperformed linear models in terms of recall and AUC, the differences were not significant, which aligns with the fact that Logistic Regression is still popular among credit risk area.

By analyzing SHAP values, we gained valuable insights into the key drivers of default risk. Features such as age, employment stability, loan amount, and spending behavior emerged as critical predictors, aligning well with domain expectations. Mature customers with stable jobs were shown to have lower default risk, whereas high spending and large borrowings were associated with increased risk.

This project also has plenty of room for future improvement. For example, limited feature set and the potential need for further hyper-parameter tuning could have constrained the MLP model's ability to leverage its complexity. More efforts will be devoted to improve the MLP model. In addition, extra models such as CNN and RNN also worth a try for credit risk predictions. These steps would ensure more robust and accurate risk predictions, ultimately helping lending institutions make more informed and reliable decisions.

References

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